

STUDY THE IMPACT ON PERSONAL EXPENDITURE USING UPI TRANSACTIONS

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I. Abstract:

In the digital era, the Unified Payment Interface (UPI) has revolutionised consumer purchasing behaviour by enabling seamless and efficient transactions. This research examines the influence of UPI transactions on personal expenditure, focusing on user adoption, spending patterns, and key driving factors. The study is based on primary data collected through a survey conducted via Google Forms. The analysis reveals a positive correlation between UPI usage and purchasing frequency, highlighting convenience and security as the primary motivators for adoption. These insights contribute to understanding the evolving digital payment landscape and its implications for consumer spending behaviour.

Keywords: UPI Transactions, Impact of UPI Transactions, UPI (Unified Payment Interface), Transaction frequency, Cashless Transaction, Digital Payments, Consumer Behaviour, Purchasing Frequency, Transaction Security.

II. Introduction:

The Unified Payments Interface (UPI) has revolutionized India's online payments market through a seamless, secure, and highly effective payment system. NPCI launched it in 2016.

Reserve Bank of India (RBI) introduced a vision statement in 2014 to develop a safe, efficient, and accessible payment system. The vision was to develop digital payments that would eliminate reliance on cash and help increase financial inclusion. The NPCI was then charged with coming up with digital payment solutions to achieve this vision, thus developing UPI.

The Rise of UPI: A Game-Changer in Digital Payments

UPI was made to be a real-time, mobile payment system where payments could be sent instantly between banks through a Virtual Payment Address (VPA), mobile number, Aadhaar number, or QR code. Contrary to NEFT and RTGS, which are subject to time limitations, UPI works 24/7 and provides unparalleled convenience.

The Impact of UPI on Consumer Spending Behaviour:

UPI has transformed consumer consumption behaviour to a large extent by facilitating transactions more quickly and conveniently. Convenience induces higher spending frequency and lower transaction avoidance, potentially influencing consumer behaviour.

1. UPI and Impulsive Purchases:

The ease of transactions through UPI can result in higher impulse spending than with the usual payment tools such as cards or cash. According to studies, when there is an ease in transactions and cash does not need to be handled physically, shoppers tend to engage in unplanned and impulsive buying.

➤ How UPI Induces Impulse Buying?

- i. Speed & Convenience: UPI shortens the time to transact by seconds, thus lessening the "cooling-off" duration in which shoppers would change their minds.
- ii. No Physical Cash Involvement: Unlike cash transactions, where consumers physically part with money, digital payments feel less tangible, making spending easier.
- iii. Seamless Integration in E-Commerce & Retail: Many online stores and apps encourage impulse buying through one-click UPI payments.

2. UPI's Influence on Consumer Expenditure Behaviour:

UPI has also played an important role in changing the way people manage their money and make payment decisions.

➤ Trends Observed in UPI Usage:

- i. Higher Frequency of Transactions: The usability of UPI results in increased transaction volumes daily. Buyers are likely to make several small transactions instead of a single large transaction.

- ii. Cash to Digital Payment Shift: UPI has promoted a behavioural change, with most individuals preferring digital payments over cash payments.
- iii. Increased Expenditure on Non-Essentials: With reduced payment barriers, users spend more on entertainment, online purchases, and food ordering services.

➤ **Consumer Confidence in UPI Transactions:**

Despite initial doubts, UPI has achieved high consumer confidence, fuelled by:

- i. Robust Security Measures: Device binding, UPI PINs, and two-factor authentication.
- ii. Support from Government & Banks: Supported by RBI & NPCI for credibility.
- iii. Refund & Dispute Settlement Mechanism: Users feel more secure because they can trace transactions and place disputes for non-payment.

According to a survey on trust in digital payments, more than 80% of Indian consumers trust the security of UPI and make it their favourite payment choice for online & offline transactions.

3. The Impact of Digital Payment Platforms on Expenditure Behaviour

Does convenience boost spending?

Yes. Studies find that contactless and digital transactions reduce the mental hurdle to paying, resulting in greater expenditure.

III. Review of literature:

1.) From Cash to Cashless: The Effect of UPI on Indian Users' Purchasing Patterns and the Development of Interfaces with Financial Responsibilities

- Dev, H., Gupta, R., Dharmavaram, S., & Kumar, D. (2024).
- The financial transaction landscape has changed due to the growing adoption of digital payments, which have made transactions safer, quicker, and easier (Dev et al., 2024). The Unified Payments Interface (UPI), one of these digital payment systems, has transformed commerce in India by making payments rapid and simple (Dharmavaram et al., 2024). However, issues with financial restraint and impulsive buying

have surfaced, leading to a study of UPI's effect on customer purchasing patterns (Gupta et al., 2024).

2.) Device Identity- Grounded stoner Authentication on Electronic Payment System for Secure E-Wallet Apps

- Hassan, M. A., & Shukur, Z. (2021).
- E-wallets have converted the payment geography, but security remains a concern. Traditional word-grounded authentication styles are vulnerable to fraud and unauthorized access (Almaiah & Al- Khasawneh, 2020). Recent studies have explored power-ground and multi-factor authentication styles, incorporating biometrics and device identity verification (Hassan & Shukur, 2021; Kumar et al., 2020).

Secure authentication fabrics combining biometric and device-grounded styles are necessary to alleviate pitfalls (Al- Khasawneh et al., 2020). Multifactor authentication using fingerprints, OTPs, and device identity has shown a pledge to enhance security without compromising stoner convenience (Kumar et al., 2020).

3.) Understanding UPI Payment Adoption Among Rural Women Perceptivity from the Technology Acceptance Model

- Swapna, N. S., & Kanthimathinathan, D.(2025).
- The relinquishment of Unified Payments Interface (UPI) payment systems among pastoral women in India remains a significant concern. Research has shown that the Technology Acceptance Model (TAM) is a precious frame for understanding the relinquishment of new technologies (Davis, 1989).

Studies have extended the TAM model to include factors similar to perceived ingeniousness and perceived threat (Venkatesh & Davis, 2000). In the environment of UPI payment relinquishment, perceived ease of use, utility, and ingeniousness have been set up to be significant predictors of relinquishment (Kumar et al., 2020).

Still, the part of the perceived threat in UPI payment relinquishment remains unclear. This study aims to probe the relinquishment of UPI payment systems among pastoral women in India, by saying the factors that impact relinquishment using an extended TAM model.

4.) Unified Payments Interface (UPI) A Digital Transformation in India

- Baliyan, D., & Singh, N. (2023).
- India's digital payment geography has experienced significant metamorphosis with the preface of Unified Payments Interface (UPI). Research has shown that UPI has simplified interbank deals, enhanced security, and contributed to the country's shift toward a cashless frugality (Reserve Bank of India, 2020). Studies have stressed the wide relinquishment of UPI, with India witnessing a significant increase in digital deals (Sahoo & Pradhan, 2020). The present study examines the elaboration, relinquishment, and impact of UPI in India, rejecting the suppositions of low relinquishment (H01) and insignificant impact (H02). The findings confirm that UPI has become the backbone of India's digital payment ecosystem, with strong interest and continued growth eventuality (Baliyan & Singh, 2023).

5.) A Study on Digital Payments System & Consumer Perception An Empirical Check

- Pandey, S. K. (2022).
- The digital payments geography in India has witnessed significant growth, with a 26.2 increase in volume during 2020- 21 (Pandey, 2022). The COVID-19 epidemic has accelerated this growth, as consumers sought to avoid cash deals due to health enterprises (Kumar et al., 2020). Research has shown that the epidemic led to a significant shift towards digital payments, with increased operation of bank transfers, mobile plutocrats, and payment cards (Sahoo & Pradhan, 2020). This shift has been driven by consumer comprehension of convenience, security, and ease of use (Pandey, 2022).

6.) A Study on UPI Transactions in India

- Kumar, S. V., & Unnisa, N. (2024).

- Unified Payments Interface (UPI) has revolutionized digital payments in India, contributing to financial inclusion, economic development, and financial literacy (Kumar & Unnisa, 2024). Research has shown that UPI has facilitated cashless payments, which can spur economic growth by reducing costs and improving efficiency (Sahoo & Pradhan, 2020).

Studies have analysed trends in UPI adoption, highlighting the growing usage of digital transactions, digital payments, and internet users in India (Kumar & Unnisa, 2024). The findings suggest that UPI has played a significant role in promoting financial inclusion and economic growth.

IV. Research Methodology:

- Objective-

- 1) To understand whether the ease of UPI payments encourages unplanned or impulsive purchases compared to more traditional forms of payment
- 2) To understand the impact of UPI on consumer expenditure behaviour. Customer trust and confidence in making financial transactions through UPI.
- 3) To examine how the ease of digital payment systems impacts impulsive spending and the frequency of transactions compared to traditional payment methods (cash or cards).

- Data Collection:

- The present study is based on primary data collected from 100 respondents from few different parts of India as well as information collected from various articles, research papers, and RBI's website. To study the impact of personal expenditure using UPI transactions, a questionnaire of 11 questions was designed to collect data from respondents through Google form.
- Sampling Unit: The sampling unit for the research was the population using UPI transactions for personal expenditure.
- Sample Size: The sample size decided in this research was 100
- Sampling Procedure: For the collection of primary data, a well-structured questionnaire was floated among different sects of the population in different parts of India.

V. Data Analysis:

For this study, primary data was collected using a structured questionnaire. The questionnaire was drafted in English.

It consisted of 11 questions, divided into two sections:

- 1.) Demographics: 3 questions to collect data on age, gender, and occupation.
- 2.) UPI Transaction Experience: 8 questions to collect data on UPI transaction User preferences, Impulse buying behaviour, spending patterns, budgeting and expense management and security perceptions.

Demographic Graphs: -

Table 1-

Age	No. of Respondents	Percentage
Below 18	0	0
18 -25	62	62
26 - 35	16	16
36 - 45	13	13
46 and above	9	9

Age Group:

100 responses

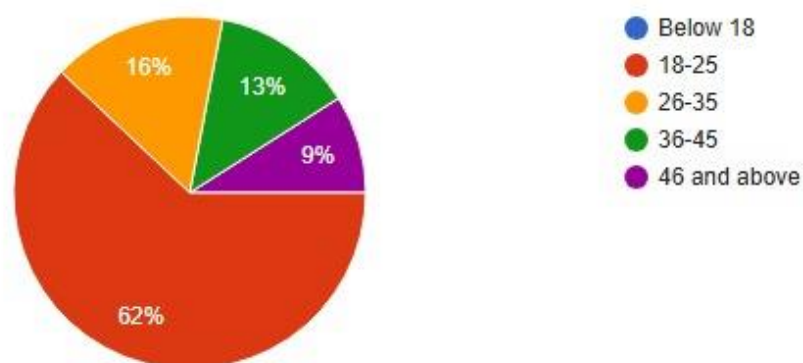


Fig 1

- Majority respondents:

Many respondents (62%) fall in the 18-25 age group, indicating that young adults are the most active users of UPI transactions.

Moderate Adoption in Other Age Groups:

- The 26-35 age group accounts for 16%, showing a moderate level of UPI usage, possibly driven by increased financial independence and digital literacy.
- The 36-45 age group makes up 13%, indicating a slightly lower adoption rate but still demonstrating a shift toward digital payments.

➤ Low Usage among Older Demographics:

Only 9% of respondents are 46 and above, reflecting slower adoption among older individuals, likely due to traditional banking preferences and lower digital literacy.

No respondents below 18, suggesting limited access or preference for cash transactions in this group.

Table 2-

Gender	No. of respondents	Percentage
Female	65	65
Male	35	35
Other	0	0

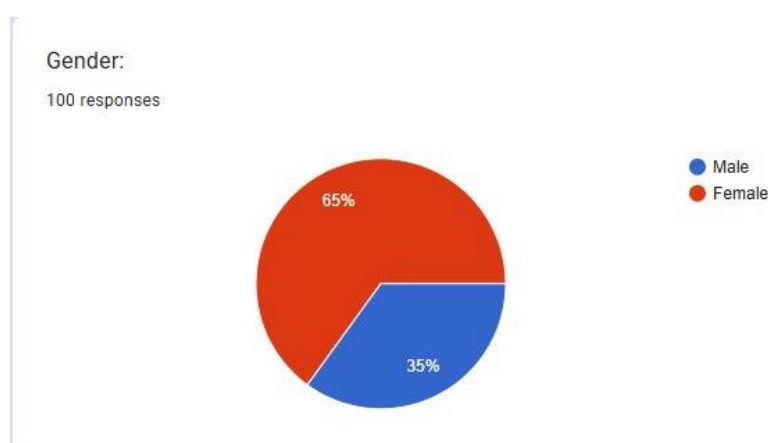


Fig 2

- Higher UPI Adoption Among Females – 65% of respondents are female, indicating greater participation in digital transactions.

- Lower Male Representation – Only 35% of respondents are male, suggesting either lower UPI usage or sample bias.
- No Respondents from Other Genders – Highlights limited data diversity in the study.

Table 3-

Occupation	No. of respondents	Percentage
Private Sector Employee	46	46
Government Employee	1	1
Self-employed/ Entrepreneur	5	5
Student	48	48

Occupation:
100 responses

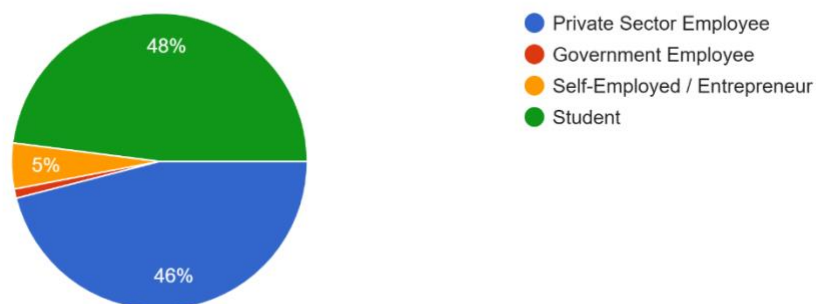


Fig 3

- Students & Private Sector Employees Are the Majority Users – Students (48%) and private sector employees (46%) dominate the respondent group, indicating that UPI usage is higher among young and working professionals.
- Minimal Participation from Government Employees & Entrepreneurs – Government employees (1%) and self-employed individuals (5%) form a small percentage, suggesting lower UPI dependency or alternative payment preferences in these groups.
- Implication on UPI-Based Expenditure – Students may use UPI for daily expenses, while private sector employees might use it for a mix of personal and professional

transactions. The low participation of government employees and entrepreneurs may indicate different spending behaviours or reliance on traditional banking.

Table 4 -

UPI Influence to buy	No. of Respondents	Percentage
Yes, often	46	46
Occasionally	32	32
Rarely	17	17
Never	5	5

Does the convenience of UPI influence your decision to buy something on impulse?
100 responses

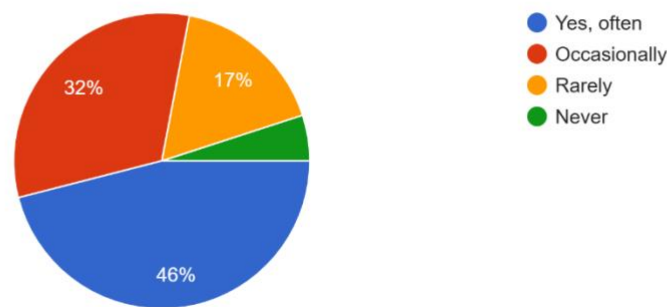


Fig 4

- **UPI Drives Impulse Purchases** – A significant 46% of respondents often make impulse purchases due to UPI’s convenience, while 32% occasionally do so. This indicates that digital payments encourage spontaneous spending.
- **Limited Financial Restraint** – Only 17% rarely and 5% never make impulse purchases, suggesting that UPI has a strong psychological impact, reducing friction in spending decisions.
- **Ease of Payment Increases Spending Behaviour** – The data suggests that the seamless, quick transaction process in UPI can lead to increased discretionary spending, potentially affecting financial discipline.

Table 5 -

Impulsive purchase	No. of Respondents	Percentage
Yes, often	39	39
Sometimes	49	49
Rarely	8	8
Never	4	4

How frequently do you find yourself making impulsive purchases when using UPI compared to cash or card payments?

100 responses

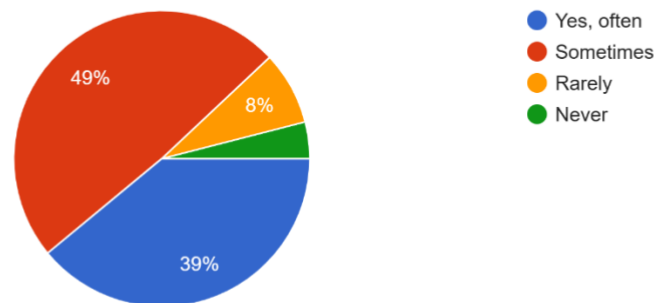


Fig 5

- Fig 5 shows Frequent Impulse Buying with UPI – A combined 88% of respondents (39% "Yes, often" + 49% "Sometimes") report making impulse purchases when using UPI, indicating that digital payments encourage unplanned spending.
- Minimal Financial Restraint – Only 8% rarely and 4% never make impulsive purchases, showing that UPI transactions reduce hesitation in spending compared to traditional payment methods.
- UPI Enhances Convenience, Leading to More Spending – The ease, speed, and seamlessness of UPI transactions contribute to higher spending tendencies, especially for impulse-driven purchases.

Table 6 -

Maintaining personal budget	No. of Respondents	Percentage
Much easier	16	16
Easier	32	32
Harder	41	41
Much harder	11	11

Do UPI transactions make it easier or harder to stick to a personal budget?

100 responses

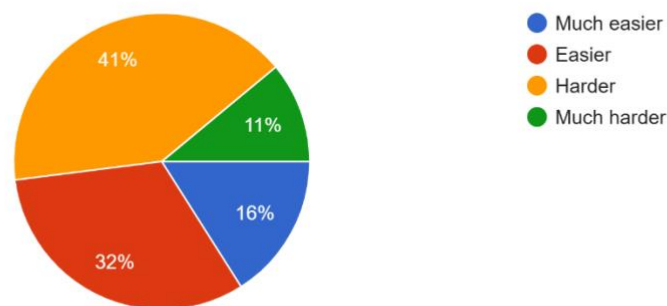


Fig 6

- Fig 6 shows Difficulty in Budget Control – 52% of respondents (41% "Harder" + 11% "Much harder") find it difficult to maintain a budget with UPI transactions, suggesting that the convenience of digital payments may lead to overspending.
- Some Benefit from UPI for Budgeting – 48% (16% "Much easier" + 32% "Easier") believe UPI helps them manage expenses, possibly due to transaction tracking and spending analytics available in banking apps.
- UPI's Impact on Financial Discipline – While UPI simplifies payments, it also encourages spending, making budgeting more challenging for many users.

Table 7 -

Spending insights	No. of Respondents	Percentage
Not at all	8	8
Slightly	29	29
Moderately	43	43
Very often	20	20

If UPI apps provided detailed spending insights (e.g., automatic categorization, spending alerts), how likely are you to use them to manage expenses?

100 responses

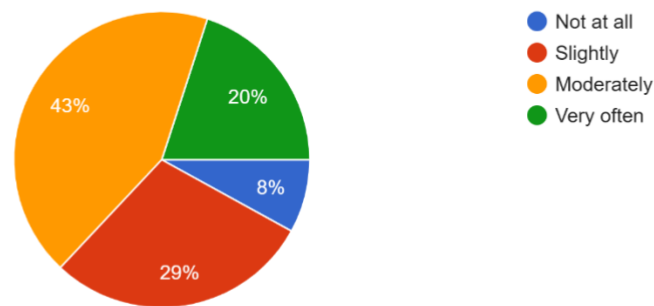


Fig 7

- Shows High Demand for Spending Insights – 63% (43% "Moderately" + 20% "Very often") of respondents are likely to use detailed spending insights provided by UPI apps, indicating strong interest in better financial tracking.
- Potential for Financial Awareness – While 8% are uninterested, 29% "Slightly" interested suggests that a significant portion of users may still benefit from awareness features if implemented effectively.
- Opportunity for UPI Apps – The data highlights the need for better spending categorization, alerts, and analytics to help users control their expenses.

Table 8 -

Secure	No. of Respondents	Percentage
Very Secure	29	29
Somewhat secure	41	41
Neutral	29	29
Not secure	0	0

How secure do you feel when making payments through UPI?

100 responses

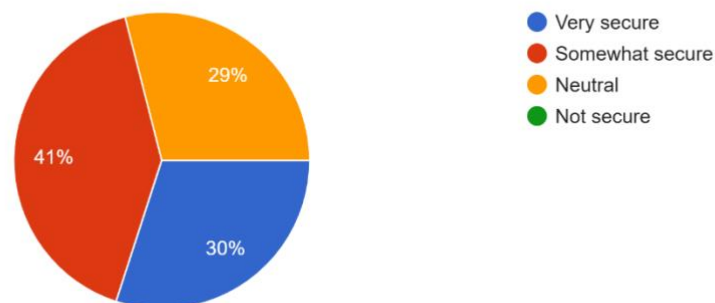


Fig 8

- Majority Feel Secure – 70% (41% "Somewhat Secure" + 29% "Very Secure") of respondents have a positive perception of UPI security, reflecting trust in the platform.
- Neutral Perception Exists – 29% of users remain neutral, indicating they neither find it secure nor insecure, possibly due to a lack of awareness about security features.
- No Strong Insecurity Concerns – 0% of respondents feel UPI is "Not Secure," reinforcing confidence in its safety mechanisms like two-factor authentication and encryption.
- Room for Improvement – While a majority feel secure, enhancing fraud awareness and reinforcing security measures could shift neutral users toward a more positive perception.

Table 9 -

Overall payment experience	No. of Respondents	Percentage
Yes	95	95
No	2	2
Unsure	3	3

Do you think UPI has improved your overall payment experience?

100 responses

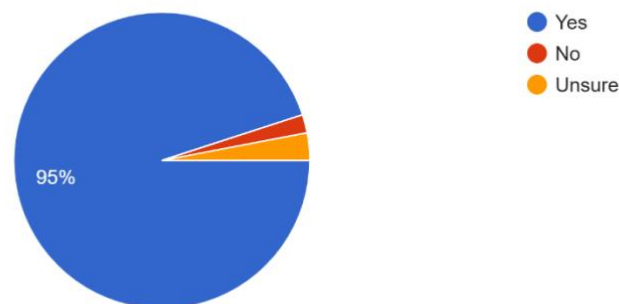


Fig 9

- This chart shows overwhelmingly Positive Response – 95% of respondents believe that UPI has improved their overall payment experience, indicating a high level of satisfaction with the platform.
- Minimal Dissatisfaction – Only 2% of users feel that UPI has not improved their payment experience, suggesting that issues like transaction failures or technical glitches are not widespread.
- Slight Uncertainty – 3% of respondents are unsure, which may imply a lack of engagement with UPI's features or a neutral stance on its impact.

Findings:

- Young adults (18-25) are the primary contributors to UPI transactions, making them the key demographic for studying the impact of personal expenditure through UPI. The adoption rate gradually declines with increasing age, suggesting a potential gap in digital payment awareness and accessibility for older generations.

Understanding spending patterns within these age groups can help in analysing how UPI impacts personal finances, budgeting habits, and financial planning.

- The study can analyse spending patterns among students and private sector employees while exploring why government employees and entrepreneurs have lower UPI usage.
- The study can explore how UPI's accessibility influences financial habits, particularly in younger or working demographics, leading to increased unplanned expenditures.
- The study can analyse how UPI's frictionless nature affects financial discipline and whether it leads to excessive discretionary spending among users.
- UPI transactions may reduce financial restraint, leading to impulsive spending and budget mismanagement. However, they also provide digital records that some users leverage for better financial control.
- Users value financial insights in UPI apps and implementing enhanced expense tracking features can improve financial discipline while maintaining the convenience of digital transactions.
- UPI transactions are widely perceived as secure, but educating neutral users about security features and fraud prevention could further boost trust and adoption.
- UPI is widely accepted as a convenient, efficient, and user-friendly payment method.
- The low percentage of negative and unsure responses reinforces UPI's role in enhancing digital payment experiences.
- Further education on advanced UPI features could convert neutral users into active adopters, maximizing its impact.

VI. Conclusion:

The Impact of UPI on Personal Expenditure

Based on the insights derived from the various survey results, UPI transactions have significantly influenced personal expenditure patterns, financial management habits, and user experience. The paper outlines both the strengths and weaknesses of UPI-based digital payment.

Key Findings & Conclusions:

1. UPI Encourages Impulse Spending:

A substantial 88% of respondents used UPI to make impulsive purchases (39% often, 49% sometimes). The seamless, quick, and cashless nature of UPI

transactions likely reduces the perceived pain of spending, making impulse buying more frequent.

2. Budgeting Becomes More Challenging:

52% respondents reported that maintaining a personal budget is more difficult or much more difficult as a result of UPI transactions. Increased convenience of digital payments can result in lower financial self-regulation, driving increasing amounts of spending at ever-increasing frequencies.

3. Spending Insights Can Improve Financial Awareness:

63% of users are likely or very likely to use UPI's spending information when it is released. This suggests that enhanced financial tracking tools, such as spending alerts and automatic categorization, could help users make more informed financial decisions.

4. UPI is Considered Secure:

70% of respondents feel secure or somewhat secure using UPI, while 29% remain neutral (none reported feeling insecure). This shows a high degree of trust in UPI security mechanisms, but it is possible to extend awareness measures to clear the neutral stance.

5. UPI Greatly Improves the Payment Experience:

Amazingly, 95% of respondents feel that UPI has made their overall payment experience better. This also highlights the more efficiency, convenience, and dependability of UPI in conducting digital payments.

Final Takeaways:

- Convenience vs. Control While UPI simplifies payments, it can lead to less conscious spending and budgeting difficulties.
- Impulsive Spending is a Concern Users tend to spend more freely with UPI, requiring better financial discipline.
- Financial insights can be a game-changer-further adoption of spending analytics and alerts can be leveraged by users to optimize spending.
- Security Trust are Strong UPI is widely perceived as safe, reinforcing its growing adoption.
- Digital Transactions Are Here to Stay the overwhelming positive feedback to UPI indicates that cash payments are going to be a relic of the past and digital payments are going to hold sway.

UPI has revolutionized personal spending habits by making transactions effortless and widely accepted. However, its ease of use comes with the risk of overspending and budget control challenges. Future developments should be directed to optimize financial logging facilities and encourage good expenditure habits to allow users to achieve the full potential of UPI without becoming financially overstretched.

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